

Is Natural Language the Right Medium for Machine Thought?

An Argument for Compressed Symbolic Reasoning in Large Language Models

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Abstract

Large language models (LLMs) reason exclusively in natural language. This convention, inherited from training on human text, is rarely questioned. We argue that it constitutes a fundamental bottleneck: natural language is an inefficient medium for structured thought. Mathematicians do not write “the sum of all integers from one to n ”—they write $\sum_{i=1}^n i$. We show that .ava notation [1], a compressed symbolic language, achieves a $2.65\times$ compression ratio over French in tokens. If reasoning scaling laws hold—more reasoning tokens imply better performance—then compressing the medium of thought could enable proportionally more reasoning steps within a fixed budget. This position paper formulates this hypothesis, gathers indirect evidence, and proposes an experimental research program to test it.

Keywords: symbolic reasoning, context compression, chain-of-thought, .ava notation, scaling laws, machine thought

tation, mathematicians wrote their reasoning in Latin prose. Al-Khwarizmi described the solution of quadratic equations in full paragraphs. The invention of symbolic notation— x , $=$, $\sum$, $\int$ —did not merely make mathematics more concise: it made possible reasoning that was *cognitively inaccessible* in prose.

We therefore ask: **Is natural language the right medium for machine thought?**

The stakes are concrete. Modern reasoning models (o1, o3, DeepSeek-R1) consume thousands, even tens of thousands of tokens for their internal chains of thought. This budget is not unlimited: the context window is finite, inference is expensive, and latency is proportional to the number of generated tokens. If the same reasoning could be expressed in 2 to 3 times fewer tokens, the implications would be considerable.

This paper is a *position paper*. We do not claim to have proven our hypothesis. We argue that it is plausible, gather converging indirect evidence, and propose a research program to test it rigorously.

1 Introduction

When a large language model “thinks,” it does so in English, French, or another human language. Chain-of-thought techniques [2] have demonstrated that verbalizing intermediate steps drastically improves performance on reasoning tasks. But no one seems to question the *medium* of that verbalization.

Consider an analogy. Before modern algebraic no-

2 The Problem: The Cost of Reasoning in Natural Language

2.1 Reasoning Scaling Laws

Recent work on reasoning scaling [3] has established a remarkable result: at fixed model size, increasing the number of tokens allocated to reasoning (compute at inference time) improves per-

formance predictably. The longer the model “thinks,” the better the result.

This has a direct consequence: the reasoning token budget is a critical resource. Every token spent on unnecessary verbosity is a token that does not contribute to effective reasoning.

2.2 The Verbosity of Natural Language

Natural language is extraordinarily verbose for expressing structured relations. Consider a simple example:

English: “If the number of elements in set A is strictly greater than the number of elements in set B, then there exists at least one element of A that does not belong to B.”

Symbolic notation: $|A| > |B| \Rightarrow \exists a \in A : a \notin B$

The prose version consumes roughly 40 tokens. The symbolic version about 15. The informational content is identical. The difference is pure linguistic redundancy: articles, prepositions, circumlocutions.

2.3 The Central Hypothesis

If an LLM could reason in a medium 2 to 3 times more compact than natural language, it could:

1. Perform 2 to 3 times more reasoning steps within the same token budget
2. Explore more branches in tree-structured reasoning
3. Maintain a richer working context within its attention window

The question is not whether compression is possible—it trivially is. The question is whether an LLM can *genuinely reason* in a compressed medium, or whether it merely translates superficially.

3 Indirect Evidence

We do not yet have direct experimental results. However, several lines of evidence converge toward the plausibility of our hypothesis.

3.1 The Mathematical Argument

The history of mathematics is itself an argument. Symbolic notation is not merely a shortcut: it structures thought. No one writes “the ratio of a circle’s circumference to its diameter” when they can write π . Leibniz’s calculus notation ($\frac{dy}{dx}$) enabled algebraic manipulations that Newton’s notation (the dot) made awkward [4].

The crucial point: notation does not merely compress the *expression* of thought—it compresses *thought itself*. A mathematician thinks in symbols, not in prose.

3.2 The Code Argument

Programmers use compact abstractions precisely because verbosity impedes reasoning. An expression like `sum(x**2 for x in A)` encodes in a few tokens what would require an entire sentence in natural language.

3.3 The .ava Notation: 2.65× Measured

In companion work [1], we designed .ava notation, a compact symbolic language for AI agent memory. Empirical measurements on a parallel corpus show a mean compression ratio of 2.65× in tokens, with peaks of 9× on the most compressible categories (scars, biases).

3.4 MetaGlyph: Symbolic Compression for Mathematics

Van Gassen [5] recently demonstrated that using extended mathematical symbols enables a 62–81% token reduction on mathematical reasoning tasks while maintaining model comprehension. This result, obtained independently, confirms that symbolic compression is not only possible but already partially effective *without dedicated fine-tuning*.

3.5 Tokenizer Bias

An often-overlooked factor: current tokenizers (BPE, SentencePiece) are trained on corpora dominated by natural language. A symbol like “ $\Rightarrow$ ” is often encoded as multiple tokens, even though it carries as much information as the word “therefore” (a single token).

A tokenizer co-designed with a symbolic notation could significantly amplify compression. We esti-

mate an additional factor of 1.5 to $2\times$ is realistic, bringing total compression to 4–5 $\times$.

4 Formal Hypothesis

We formulate the following hypothesis:

H1: *An LLM fine-tuned to reason in compressed symbolic notation can solve problems requiring more reasoning steps than the same model reasoning in natural language, at a fixed token budget.*

More formally: let M be a language model, M' the same model fine-tuned on reasoning chains in symbolic notation, T a maximum token budget, and $D(k)$ the set of problems requiring at least k reasoning steps. We predict:

$$\exists k^* : \text{Acc}(M', D(k^*), T) > \text{Acc}(M, D(k^*), T) \quad (1)$$

where $\text{Acc}(M, D, T)$ is the accuracy of model M on problems D with a budget of T tokens, and k^* is large enough that the budget T becomes the limiting factor for M but not for M' .

The corresponding null hypothesis is that compression only degrades reasoning quality, even if it increases reasoning quantity.

5 Proposed Research Program

5.1 Exp. 1: Chain-of-Thought in .ava vs Natural Language

Protocol. Fine-tune an LLM (e.g., Llama 3, Qwen 2.5) on pairs of (problem, reasoning in .ava) and (problem, reasoning in English). Evaluate on multi-step reasoning benchmarks (GSM8K, MATH, ARC-Challenge) at a fixed token budget.

Primary metric. Accuracy on problems requiring $> k$ steps, for increasing values of k .

Crucial control. Verify that the .ava model does not merely “translate” its reasoning but genuinely reasons in the compressed notation.

5.2 Exp. 2: Reasoning Depth at Fixed Budget

Protocol. Measure the number of explicit reasoning steps produced by the model at a fixed token

budget (e.g., 2048 tokens). Compare the depth achieved in .ava vs natural language.

Prediction. The .ava model should achieve $\sim 2.5\times$ more reasoning steps at the same budget.

5.3 Exp. 3: Tokenizer + Notation Co-Design

Protocol. Train a BPE tokenizer on a mixed corpus including a significant proportion of .ava notation. Measure the impact on compression ratio and downstream performance.

5.4 Exp. 4: Latent Representation Analysis

Protocol. Compare hidden states of a model reasoning in .ava vs natural language on identical problems. Use linear probes to determine whether internal representations differ qualitatively.

Question. Does the model “think differently” in .ava, or does it merely encode the same representations in a different surface format?

6 Related Work

Chain-of-thought prompting. Wei et al. [2] showed that explicit reasoning improves LLM performance. Our work does not contradict this result: we question the *format* of reasoning, not its utility.

Thinking tokens and internal reasoning. Models o1 [6] and DeepSeek-R1 [7] use “thinking tokens” in natural language. Coconut [8] proposes reasoning in continuous latent space, bypassing the textual format entirely. Our approach is intermediate: a textual format, but symbolic and compressed.

Compressed reasoning. Munkhbat et al. [9] propose self-compressed chain-of-thought, where the model learns to condense its reasoning. Our work is complementary but distinct: we propose changing the *language* of reasoning, not merely its length.

Symbolic AI. The symbolic AI tradition (GOFAI) used logical formalisms for decades. Our proposal differs: we do not propose replacing LLMs with symbolic systems, but equipping LLMs with a more efficient medium of thought.

Symbolic compression. MetaGlyph [5] shows

that compact mathematical symbols reduce tokens by 62–81% on mathematical tasks. Our approach generalizes this principle beyond mathematics alone.

7 Risks and Objections

We acknowledge several serious objections to our hypothesis.

Translation vs reasoning. The most fundamental risk: a model fine-tuned on .ava might learn to *translate* its internal reasoning (in natural language) to .ava, rather than reasoning directly in .ava. Exp. 4 (hidden state analysis) is specifically designed to detect this scenario.

Loss of nuance. Compression eliminates redundancy, but natural language redundancy is not always useless. It aids disambiguation, context maintenance, and error correction.

Evaluation. Evaluating reasoning in symbolic notation is more difficult than in natural language. Human evaluators and judge models will need adaptation.

Generalization. Symbolic compression is natural for mathematics and logic. It is less natural for commonsense reasoning. Our hypothesis may hold only for certain problem classes.

Training bias. Current LLMs have been massively trained on natural language. Their ability to reason in symbolic notation is inherently limited by this history. Fine-tuning may not suffice: partial pre-training may be necessary.

8 Conclusion

We have posed a question we consider fundamental: is natural language the right medium for machine thought? We have argued that it is not, drawing on the historical analogy with mathematics, on empirical compression measurements ($2.65\times$ for .ava), and on the logic of reasoning scaling laws.

Our hypothesis is speculative. It could be wrong—natural language might be, for reasons we do not yet understand, the optimal format for neural network reasoning. But if it is right, the implications are profound: we could significantly increase the reasoning capacity of LLMs without increasing their size or inference cost, simply by changing the language in which they think.

The question deserves to be asked. The research program we propose is designed to answer it.

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